

Ahfad University for Women
School of Management Studies
Management Information Systems

Simple Prototype and System Requirements : MedAccess Sudan

Students' names and IDs:

1. Lina Ibrahim Rahamtalla Hamad — 201800182
2. Ola Osama Awad Musa — 201800470
3. Suhandha Faisal Mohamed Said — 201800196

1. Project Overview.....	2
2. Functional Requirements.....	2
3. Non-Functional Requirements.....	9
4. Simple Prototype.....	17
4.1 Prototype Overview.....	17
4.2 Prototype Objective.....	18
4.3 Prototype Screens.....	18
4.4 Prototype Flow.....	20
4.5 Prototype Navigation Flow.....	21
4.6 Prototype Scope.....	21
4.7 Prototype Design Approach.....	21
4.8 Prototype Tool.....	22
4.9 Interactive Prototype.....	22

1. Project Overview

MedAccess Sudan is an integrated digital healthcare platform designed to improve access to healthcare services in Sudan. The system will connect patients with healthcare providers through both a web platform and a mobile application.

The platform will allow users to search for healthcare providers, book appointments, schedule home visits, reserve laboratory tests, receive appointment updates, and manage medical information.

The system will also include an administrative dashboard for healthcare-provider management and will support future teleconsultation functionality.

2. Functional Requirements

Functional requirements define the operations and services that the MedAccess Sudan system shall provide to its users. The requirements below are derived directly from the functions specified in the project proposal and are written so that each requirement can later be implemented and tested individually.

ID	Functional Requirement	Detailed Description	Main Actor(s)	Expected Output / Result	Phase
FR-01	Patient Registration	The system shall allow a new patient to create an account by entering the required registration information. The system shall validate the submitted information and create a patient record after successful registration.	Patient	A new patient account is created and stored in the system.	Current

FR-02	User Authentication	The system shall allow registered users to log in using valid authentication credentials. The system shall verify the submitted credentials before granting access to protected system functions.	Patient, Administrator	The authenticated user is granted access to the appropriate system interface.	Current
FR-03	Healthcare Provider Record Management	The system shall maintain healthcare-provider records containing the information required for patients to identify and select providers, including provider name, specialization, location, and availability information.	Administrator	Healthcare-provider information is stored and made available to the relevant system functions.	Current
FR-04	Healthcare Provider Verification and Administration	The system shall allow an administrator to manage and verify healthcare-provider records through the administrative dashboard. This shall include adding, reviewing, updating, and managing provider	Administrator	Healthcare-provider records are maintained and verified for use within the system.	Current

		information available on the platform.			
FR-05	Healthcare Provider Search	The system shall allow patients to search for healthcare providers available through the MedAccess Sudan platform.	Patient	A list of healthcare providers matching the search is displayed.	Current
FR-06	Healthcare Provider Filtering	The system shall allow patients to filter healthcare providers according to specialization, location, and availability . The system shall return providers that satisfy the selected criteria.	Patient	Filtered healthcare-provider results are displayed.	Current
FR-07	Healthcare Provider Information Display	The system shall allow patients to view relevant information about a selected healthcare provider, including specialization, location, and available appointment options.	Patient	The selected provider's details and availability are displayed.	Current

FR-08	Appointment Availability Display	The system shall display the available appointment dates and/or time slots associated with the selected healthcare provider so that the patient can choose an appropriate appointment.	Patient	Available appointment options are presented to the patient.	Current
FR-09	Appointment Booking	The system shall allow a patient to select an available healthcare provider, date, and appointment time and submit an appointment-booking request. The system shall record the confirmed appointment.	Patient	A new appointment record is created.	Current
FR-10	Appointment Management	The system shall allow patients to access and manage their appointment information after booking. This shall include viewing their scheduled appointments and their current appointment information or status.	Patient	The patient's appointment information is displayed and maintained by the system.	Current

FR-11	Home-Visit Scheduling	The system shall allow patients to request and schedule a home visit as one of the healthcare services provided through the platform. The system shall record the required scheduling information for the request.	Patient	A home-visit request or scheduled home visit is recorded.	Current
FR-12	Laboratory Test Reservation	The system shall allow patients to select and reserve laboratory-testing services through the platform and record the relevant reservation information.	Patient	A laboratory-test reservation is created and stored.	Current
FR-13	Appointment Updates	The system shall provide patients with updates concerning their appointments when relevant appointment information changes.	Patient	Updated appointment information is presented to the patient.	Current
FR-14	Patient Notifications	The system shall provide notifications related to appointments and relevant scheduling information	Patient	The patient receives an appointment-related notification or update.	Current

		through the mobile application.			
FR-15	Patient Medical History Management	The system shall maintain patient medical-history information and allow the relevant patient medical-history records to be stored and accessed through the system.	Patient / Authorized System User	Patient medical-history information is stored and can be retrieved when required.	Current
FR-16	User Data Management	The system shall store and retrieve registered-user information required for account operation and use of healthcare services.	System	User information is stored and available to authorized system functions.	Current
FR-17	Appointment Data Management	The system shall store and retrieve appointment information, including the patient, healthcare provider, scheduled date/time, and relevant appointment details.	System	Appointment records are maintained in the database.	Current

FR-18	Web and Mobile Data Synchronization	The mobile application shall communicate with the backend through API integration. Changes to relevant records, including patient, healthcare-provider, and appointment information, shall be synchronized so that the web platform and mobile application use consistent backend data.	Web Application, Mobile Application, Backend	Consistent information is available through both interfaces.	Current
FR-19	Administrative Dashboard	The system shall provide an administrative interface through which authorized administrators can access healthcare-provider management functions.	Administrator	The administrator can access and perform provider-management activities.	Current
FR-20	Teleconsultation	The system shall support remote consultations between patients and healthcare providers using video and/or audio communication when the teleconsultation module is implemented.	Patient, Healthcare Provider	A remote consultation session can be conducted through the platform.	Future Phase

3. Non-Functional Requirements

Non-functional requirements define the quality attributes, technical constraints, and operational standards that MedAccess Sudan shall satisfy while performing its functional requirements.

ID	Category	Non-Functional Requirement	Detailed Description	Implementation Consideration	Verification
NFR-01	Usability	The system shall provide a simple and user-friendly interface.	Patients shall be able to navigate the platform and access major healthcare services without unnecessary complexity. Interfaces should use understandable labels, consistent navigation, and clear actions.	Use consistent menus, buttons, forms, messages, and page layouts across the system.	Conduct usability testing by asking representative users to complete tasks such as finding a doctor and booking an appointment.
NFR-02	Accessibility	The system shall provide an accessible interface with readable and understandable content.	Text, navigation controls, forms, and system messages shall be presented clearly to support users with different levels of digital experience.	Use readable font sizes, descriptive form labels, adequate contrast, keyboard-accessible controls where applicable, and clear validation messages.	Review interfaces against accessibility guidelines and test important screens manually.

NFR-03	Responsive Design	The web application shall adapt to different screen sizes.	The website shall remain usable when accessed from desktop computers, tablets, and mobile devices.	Implement responsive layouts using CSS media queries, flexible containers, and responsive components.	Test the website at multiple viewport sizes and on different device types.
NFR-04	Cross-Platform Compatibility	The mobile application shall support Android and iOS platforms.	The React Native application shall provide the required MedAccess Sudan functionality on both supported mobile operating systems.	Use cross-platform React Native components where possible and isolate platform-specific code when necessary.	Build and test the application on Android and iOS environments.
NFR-05	Web Compatibility	The web platform shall operate correctly on commonly used modern web browsers.	Core MedAccess Sudan functions shall remain usable across supported browsers.	Use standards-based HTML, CSS, and JavaScript and avoid unnecessary browser-specific features.	Test major workflows on multiple modern browsers.
NFR-06	Authentication Security	The system shall securely authenticate users before allowing access to protected system functions and information.	Protected patient and administrative functions shall only be available after successful authentication.	Use secure login sessions or tokens, password hashing, access controls, and protected routes/API endpoints.	Attempt authenticated and unauthenticated access to protected functions and confirm unauthorized requests are rejected.

NFR-07	Authorizati on	The system shall restrict functions and data according to authorized user roles.	Patients and administrators shall only access system functions and information appropriate to their roles. Additional healthcare-provider permissions can be incorporated if provider accounts are later included.	Implement role-based access control in both frontend navigation and backend authorization.	Test each role against permitted and prohibited actions.
NFR-08	Data Privacy	Patient and healthcare information shall be protected from unauthorized access.	Personal information, medical history, appointment details, and other sensitive healthcare data shall only be accessible to authorized users and system components.	Apply authentication, authorization, secure API communication, and database access restrictions.	Conduct access-control tests and verify that unauthorized accounts cannot retrieve protected records.

NFR-09	Secure Data Storage	Sensitive application data shall be stored securely.	User records, medical information, provider information, and appointment data stored in MySQL shall be protected from unauthorized access and inappropriate exposure.	Hash passwords rather than storing plain text; use restricted database accounts; validate database queries and protect sensitive fields.	Inspect database configuration and verify that passwords and sensitive authentication information are not stored in plain text.
NFR-10	Secure Communication	Communication between system components shall protect information while it is transmitted.	Data exchanged between the website, mobile application, APIs, and backend shall use secure communication mechanisms.	Use HTTPS/TLS for deployed web and API communication. Avoid transmitting credentials or sensitive records through unsecured channels.	Verify deployed endpoints use HTTPS and reject inappropriate insecure access where applicable.
NFR-11	Input Validation	The system shall validate user-supplied data before processing or storing it.	Registration information, appointment requests, search parameters, medical records, and administrative inputs shall be checked for validity.	Perform validation on both the client side for usability and the server side for security and data integrity.	Submit invalid, missing, malformed, and unexpected input and verify that it is rejected appropriately.

NFR-12	Data Integrity	The system shall maintain accurate and consistent data.	Patient, provider, appointment, home-visit, laboratory, and medical-history records shall remain internally consistent when records are created or modified.	Use database constraints, appropriate data types, primary/foreign keys, validation, and transactions where required.	Attempt invalid database relationships or conflicting operations and verify that invalid states are prevented.
NFR-13	Reliability	The system shall perform its supported functions consistently without corrupting or losing valid information.	Successful operations such as account creation and appointment booking shall produce predictable results and preserve the corresponding records.	Add error handling, database transaction management, and clear failure responses.	Repeat major system workflows and test failure scenarios such as invalid requests and database errors.
NFR-14	Error Handling	The system shall handle errors without exposing sensitive technical information or leaving data in an inconsistent state.	Users shall receive understandable error messages when an operation fails, while internal technical details remain protected.	Implement centralized backend error handling and user-friendly frontend messages.	Trigger known failure conditions and review both the user-facing response and system behavior.

NFR-15	Performance	The system shall respond to ordinary user operations within an acceptable period under normal usage conditions.	Search, authentication, appointment retrieval, and common page operations should not introduce unnecessary delays.	Optimize database queries, API calls, frontend assets, and indexes as the system grows.	Measure response times for representative operations during testing.
NFR-16	Scalability	The system architecture shall support future expansion without requiring complete redevelopment.	MedAccess Sudan shall be capable of accommodating additional users, healthcare providers, locations, services, and future modules such as payments and insurance.	Use modular backend services, normalized database structures, APIs, reusable components, and configurable service modules.	Add sample records or modules during testing and confirm that core architecture does not require redesign.
NFR-17	Maintainability	The system shall use a modular architecture that supports future modification and maintenance.	Individual modules should be capable of being updated with minimal unnecessary impact on unrelated functionality.	Separate frontend, backend, database, API, authentication, appointment, provider, laboratory, and other logical components. Follow consistent naming and code organization.	Review the project structure and test whether modules can be modified independently.

NFR-18	Code Quality	The software shall be organized and documented sufficiently to support continued development.	Source code should follow consistent conventions and provide meaningful names and documentation where needed.	Establish folder structure, naming conventions, reusable functions/components, comments for complex logic, and version control.	Conduct code review and inspect consistency of the repository.
NFR-19	API Consistency	The web and mobile applications shall interact with the backend through consistently defined APIs.	API behavior, request structures, responses, authentication, and error handling should remain consistent across clients.	Define API routes and standard JSON request/response structures before implementing multiple clients.	Test the same backend operations from web/mobile clients or through an API-testing tool.
NFR-20	Data Synchronization	Relevant information accessed through the web and mobile applications shall remain synchronized through the shared backend.	Updates to users, healthcare providers, appointments, and other shared records shall be reflected consistently across supported interfaces.	Use one authoritative backend/database rather than maintaining unrelated data stores for web and mobile applications.	Modify data through one interface and verify that the updated information can subsequently be retrieved through the other.

NFR-21	Extensibility	The architecture shall allow future features to be incorporated without rewriting the complete system.	Planned future functionality, particularly teleconsultation, payment integration, and insurance-related capabilities, should be capable of being added as separate modules.	Design clear service boundaries and avoid tightly coupling future services to unrelated components.	Prototype the addition of a new service/module and assess whether existing core modules require major modification.
NFR-22	Database Maintainability	The database shall use a structured design capable of supporting the system's current and future records.	Tables and relationships shall clearly represent patients, healthcare providers, appointments, medical information, and other implemented services.	Use primary keys, foreign keys, indexes, appropriate data types, constraints, and normalized structures where appropriate.	Review the ERD/database schema and test referential integrity.
NFR-23	Backup and Recovery	Application data shall be capable of being backed up and restored.	The system should provide a mechanism for recovering important records if database data is lost or corrupted.	Establish MySQL backup procedures and document a restoration process before production deployment.	Restore a test backup and verify that expected records are recovered correctly.

NFR-24	Auditability	Important administrative and sensitive record changes should be traceable where appropriate.	The system should preserve enough information to identify significant administrative actions and modifications to sensitive healthcare records.	Maintain appropriate timestamps and, for important changes, identify the authenticated user responsible for the action.	Modify selected administrative records and verify that the relevant timestamps/action records are available.
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4. Simple Prototype

4.1 Prototype Overview

A clickable mobile application prototype was developed for **MedAccess Sudan** to demonstrate one of the main functions of the proposed healthcare platform. The prototype focuses on the patient journey of **searching for a healthcare provider and booking an appointment**.

The purpose of the prototype is to provide a visual and interactive representation of how a patient would interact with MedAccess Sudan before the complete system is developed. It demonstrates the sequence of actions required to access the platform, search for an appropriate healthcare provider, view provider information and availability, select an appointment, confirm the booking, and receive confirmation.

The prototype was designed using **Figma** as a mobile application interface. It consists of eight connected screens and uses interactive navigation to simulate the appointment-booking process.

Although the complete MedAccess Sudan system will include additional functions such as home-visit scheduling, laboratory test reservations, appointment management, notifications, and patient medical-history management, these functions are outside the detailed interactive scope of this prototype. They are represented where appropriate in the interface, while the prototype concentrates on demonstrating the main appointment-booking workflow.

4.2 Prototype Objective

The main objective of the prototype is to simulate how a patient can use MedAccess Sudan to find a suitable healthcare provider and successfully book an appointment.

The prototype demonstrates the following process:

Patient Login → Patient Home → Find a Doctor → Search and Filter Healthcare Providers → View Search Results → View Doctor Profile → Select Appointment → Confirm Booking → Booking Successful

The prototype also demonstrates basic usability principles including clear navigation, consistent interface components, visible system feedback, back navigation, and clear confirmation of important user actions.

4.3 Prototype Screens

The prototype consists of the following eight screens:

Screen 1 – Login

The Login screen represents the entry point to the MedAccess Sudan application. It displays the MedAccess Sudan branding and provides fields for the patient's email address or phone number and password. It also contains a **Log In** button and a **Create Account** option for new users.

For prototype purposes, selecting Log In simulates successful authentication and directs the patient to the Home screen.

Screen 2 – Patient Home

The Home screen provides the patient with access to the main healthcare services offered by MedAccess Sudan. It displays a patient greeting, notification access, and four main service options:

- Find a Doctor
- Home Visit
- Laboratory Tests
- My Appointments

The prototype uses **Find a Doctor** as the starting point of the main interactive workflow. The remaining options represent other functions planned for the complete MedAccess Sudan system.

Screen 3 – Find a Doctor

The Find a Doctor screen enables the patient to define criteria for finding an appropriate healthcare provider.

The patient can filter healthcare providers according to:

- Specialization
- Location
- Availability

After selecting the required criteria, the patient selects **Search** to view matching healthcare providers.

Screen 4 – Search Results

The Search Results screen displays healthcare providers matching the selected search criteria.

Each provider result presents relevant information such as the healthcare provider's name, specialization, location, and availability. The prototype uses fictional healthcare-provider information for demonstration purposes.

The patient can select a provider from the results to view additional information.

Screen 5 – Doctor Profile

The Doctor Profile screen provides additional information about the selected healthcare provider. It displays the provider's name, specialization, location, professional information, and availability.

From this screen, the patient can proceed to the appointment-booking process by selecting the **Book Appointment** option.

Screen 6 – Select Appointment

The Select Appointment screen allows the patient to choose an available appointment date and time.

Available, selected, and unavailable appointment slots are visually distinguished to make the selection process clear.

After selecting an appropriate date and time, the patient proceeds to review the appointment.

Screen 7 – Confirm Booking

The Confirm Booking screen allows the patient to review the selected appointment before completing the booking.

The appointment summary displays relevant information including:

- Patient
- Healthcare provider
- Appointment date
- Appointment time
- Location

The patient selects **Confirm Booking** to complete the appointment-booking process.

Screen 8 – Booking Successful

The final screen provides feedback that the appointment has been successfully booked.

The screen displays a successful booking confirmation and the relevant appointment information. It also provides options to **View Appointments** or return **Back to Home**.

This screen represents the successful completion of the main prototype workflow.

4.4 Prototype Flow

The prototype follows the sequence below:

1. Patient Login

The patient enters login information and accesses the application.

↓

2. Patient Home

The patient views the available MedAccess Sudan services and selects **Find a Doctor**.

↓

3. Doctor Search

The patient selects the required specialization, location, and availability.

↓

4. Search Results

The system displays healthcare providers matching the selected criteria.

↓

5. Doctor Profile

The patient selects a healthcare provider and reviews the provider's information.

↓

6. Appointment Selection

The patient selects an available appointment date and time.

↓

7. Booking Confirmation

The patient reviews the appointment information and selects **Confirm Booking**.

↓

8. Booking Successful

The system confirms that the appointment has been successfully booked.

4.5 Prototype Navigation Flow

The primary navigation path is:

Login → Home → Find Doctor → Search Results → Doctor Profile → Select Appointment → Confirm Booking → Booking Successful

Back-navigation controls allow the patient to return to the previous screen during the booking process.

After successfully completing the booking, the patient can return to the Home screen or proceed to the appointments area.

4.6 Prototype Scope

The prototype is intentionally limited to the **patient appointment-booking process** and does not represent the complete implementation of MedAccess Sudan.

The following functions are represented or planned within the wider system but are not fully simulated in this prototype:

- Home-visit scheduling
- Laboratory test reservations
- Appointment management
- Patient notifications
- Patient medical-history management
- Administrative healthcare-provider management
- Web and mobile data synchronization
- Future teleconsultation functionality

These functions will form part of the wider system development according to the project's functional requirements.

The prototype therefore serves as an early visual representation of the proposed system rather than a fully developed healthcare application.

4.7 Prototype Design Approach

The MedAccess Sudan prototype follows a clean and consistent mobile healthcare interface. The design prioritizes simplicity, readability, clear navigation, and ease of use.

Consistent interface elements are used throughout the prototype, including buttons, cards, input fields, navigation controls, appointment slots, typography, spacing, and visual hierarchy.

The prototype also provides visual feedback for different appointment states, including available, selected, and unavailable appointment slots. Interactive transitions between screens simulate how the final mobile application may behave.

4.8 Prototype Tool

The prototype was created using **Figma**. Interactive connections between the screens were used to simulate the patient journey and allow the main appointment-booking workflow to be demonstrated without implementing the complete frontend, backend, database, or API infrastructure.

The Figma prototype will serve as a visual reference during later system development and can guide the implementation of the mobile application interface.

4.9 Interactive Prototype

The interactive prototype can be accessed through the following link:

Figma Prototype Link:

<https://www.figma.com/proto/q1AWnsVcmYJxBZfm7Ff63G/Untitled?node-id=0-1&t=dSqlWwm4rT6X5W3m-1>

Figure 1: Login Screen

9:41

MedAccess Sudan

Connecting Sudan with healthcare experts

Email or Phone Number

e.g. amira@gmail.com or +249...

Password

Enter password

[Forgot Password?](#)

Log In

Don't have an account? [Create Account](#)

Figure 2: Patient Home Screen

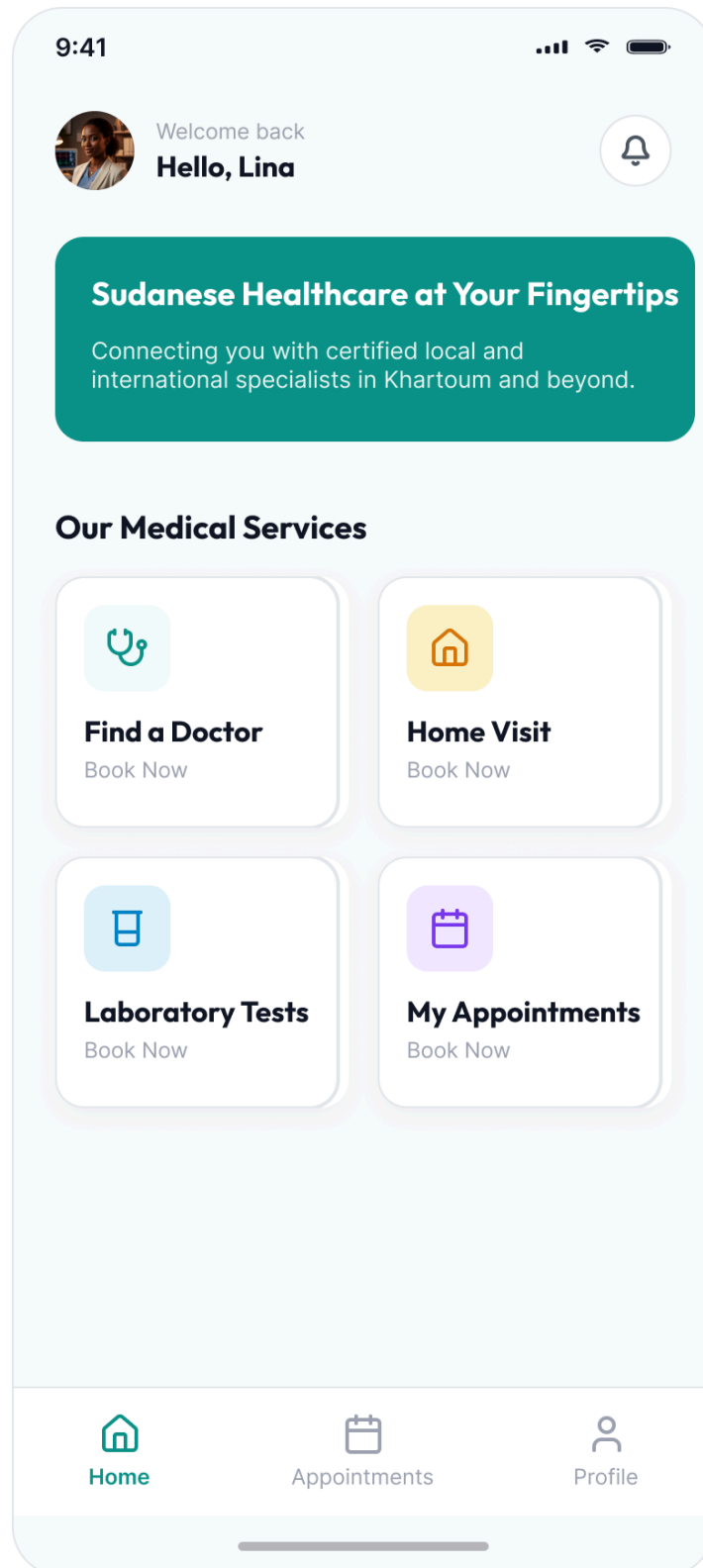
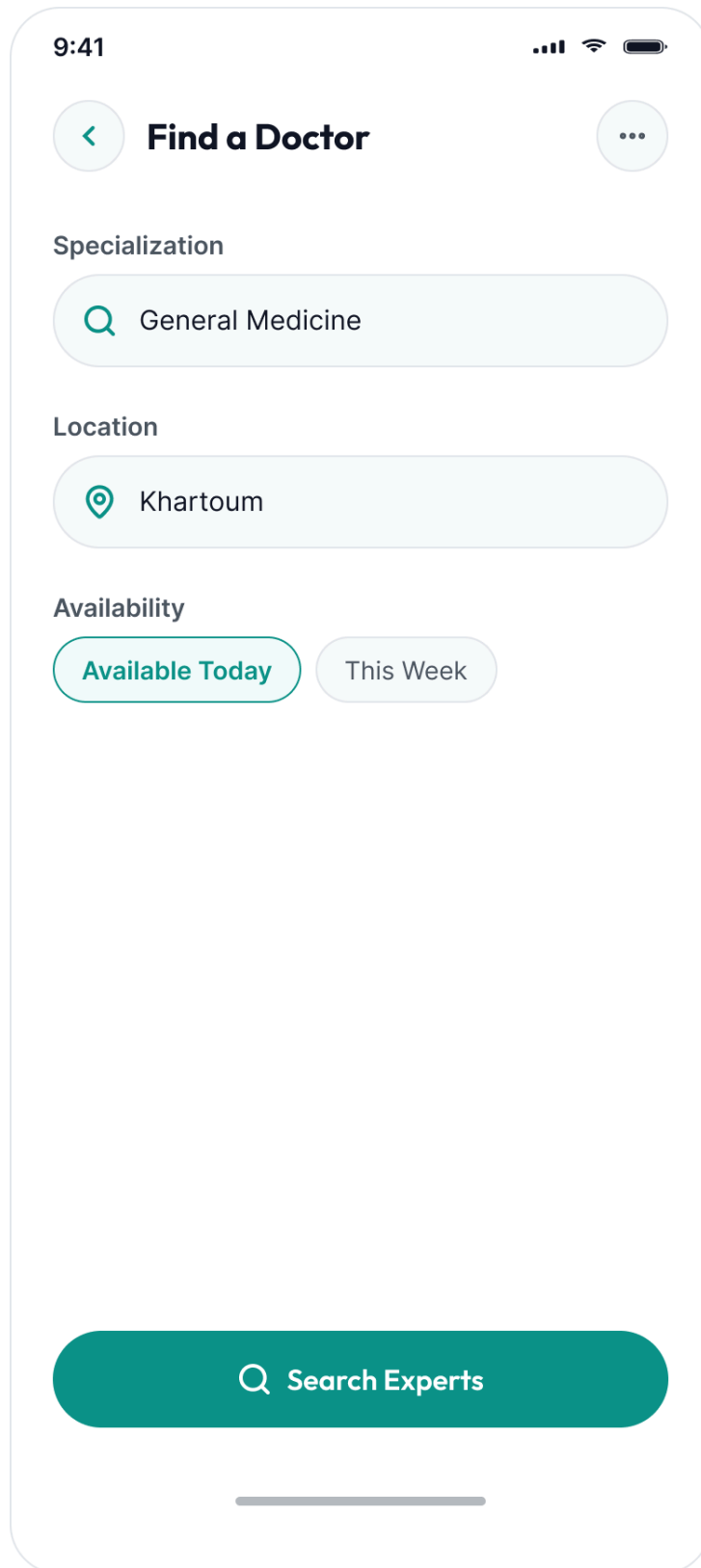


Figure 3: Find a Doctor Screen



The image shows a mobile application screen titled "Find a Doctor". At the top, the status bar displays the time 9:41, signal strength, Wi-Fi, and battery icons. Below the status bar is a header with a back arrow, the title "Find a Doctor", and a menu icon. The main content area is divided into three sections: "Specialization" with a search bar containing "General Medicine", "Location" with a search bar containing "Khartoum", and "Availability" with two buttons: "Available Today" (highlighted in teal) and "This Week". At the bottom, there is a large teal button with a magnifying glass icon and the text "Search Experts".

9:41

< Find a Doctor

Specialization

Q General Medicine

Location

Q Khartoum

Availability

Available Today This Week

Q Search Experts

Figure 4: Healthcare Provider Search Results

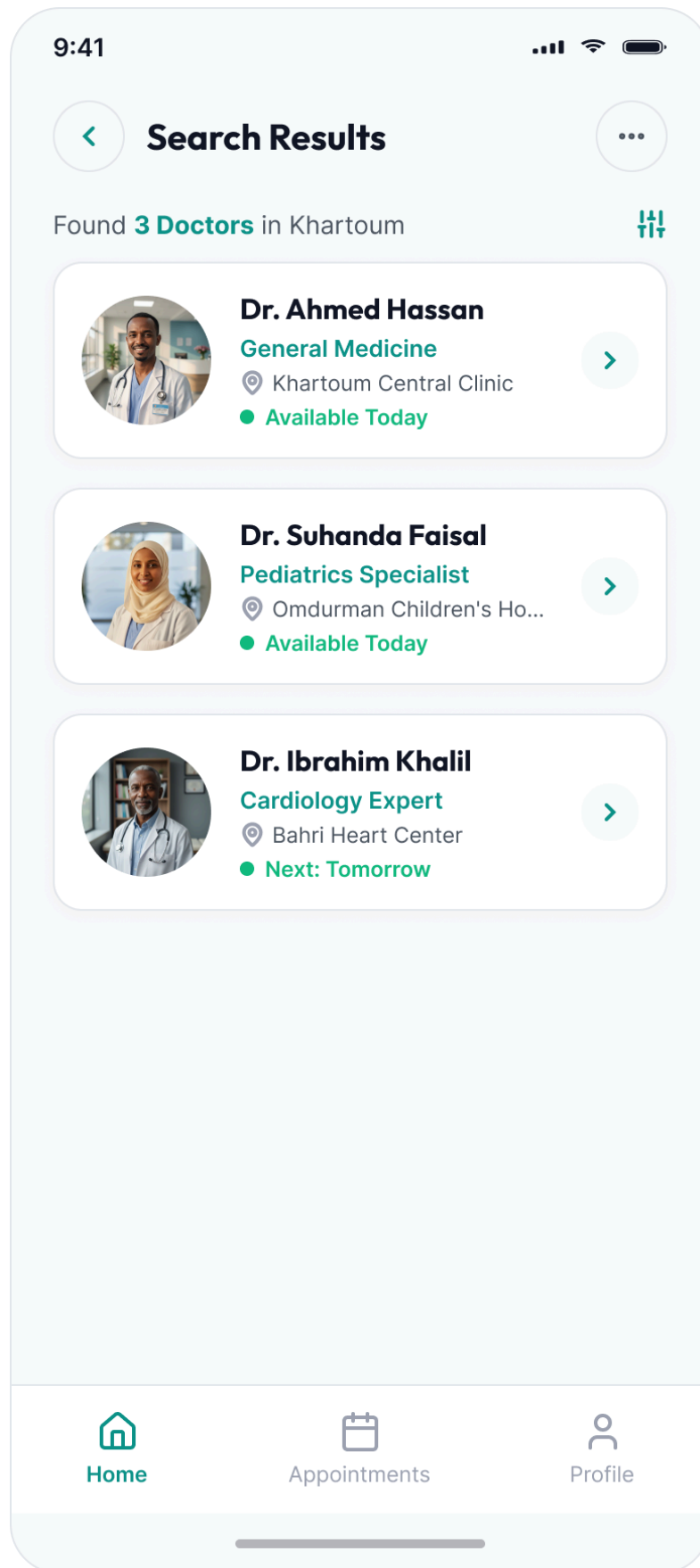


Figure 5: Doctor Profile Screen

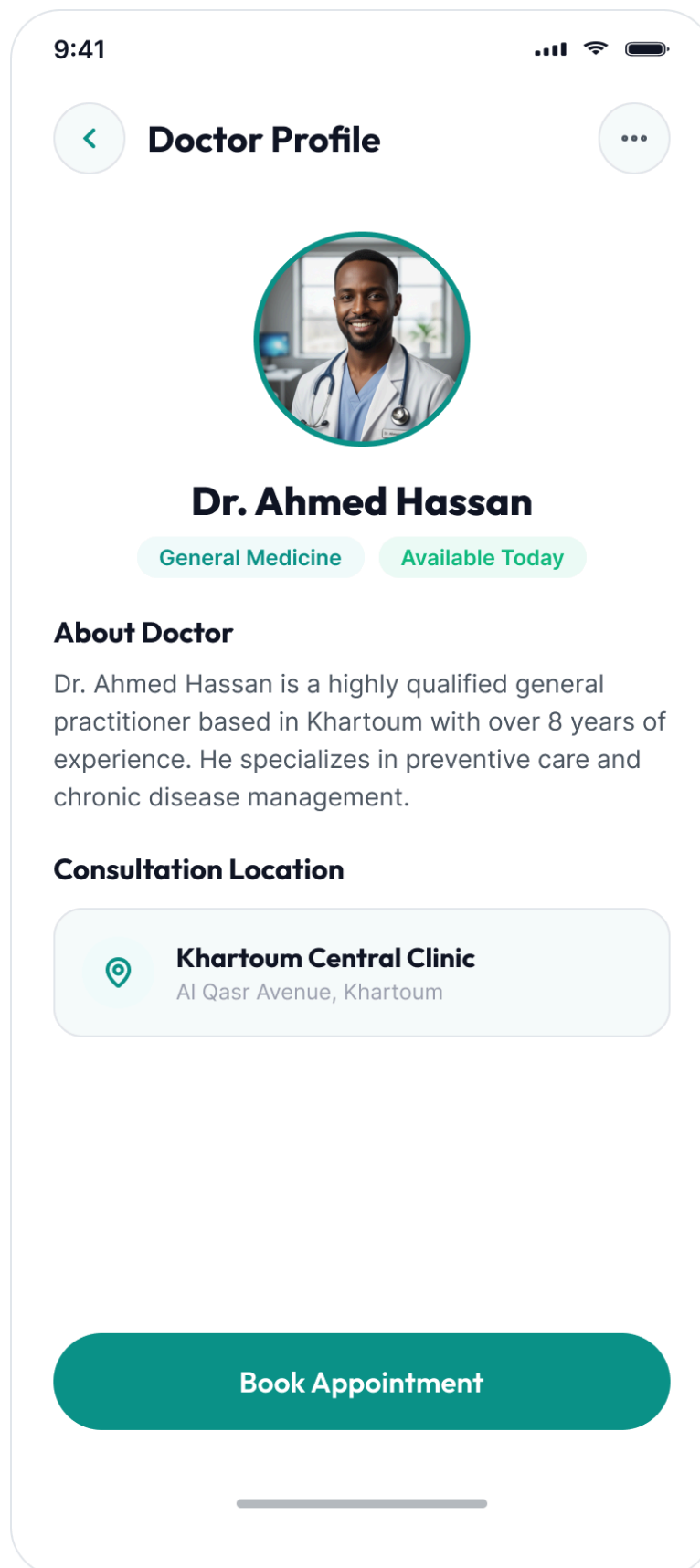
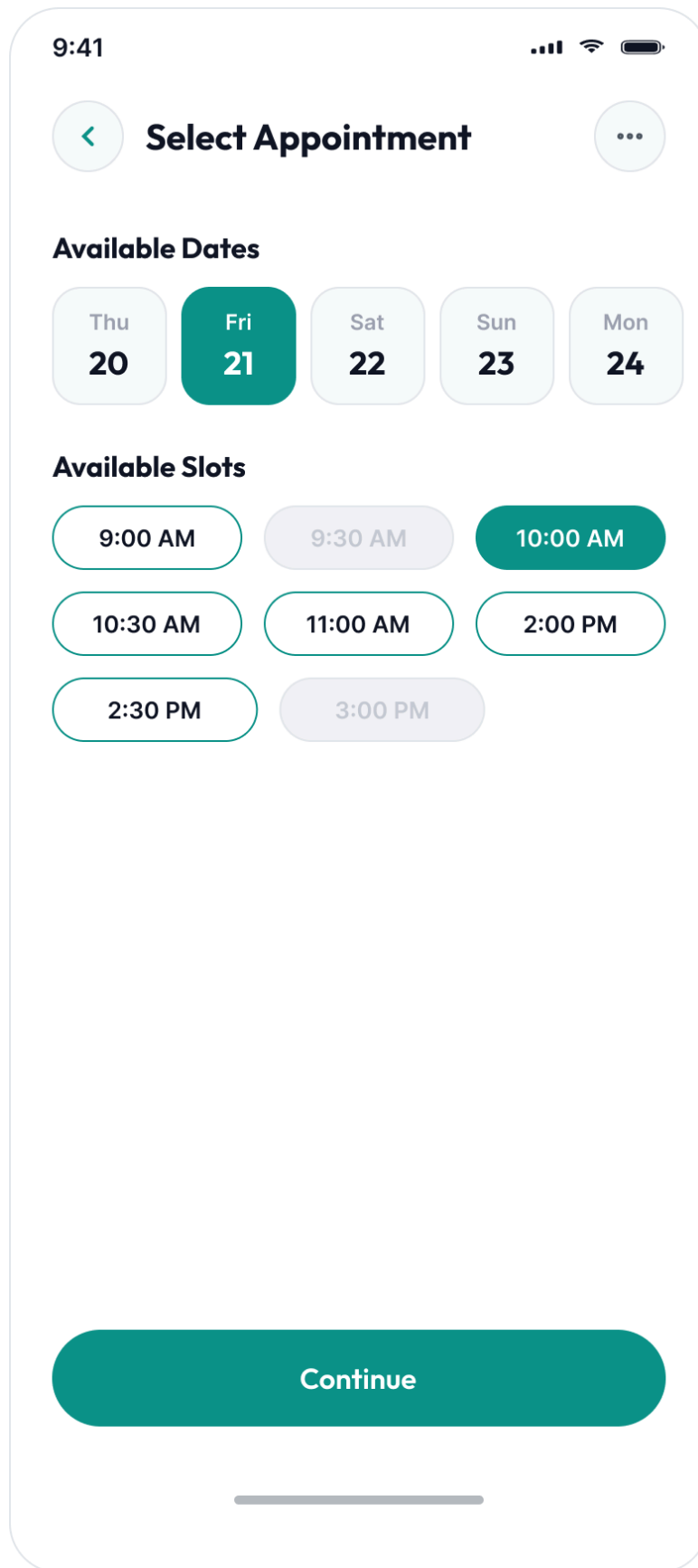


Figure 6: Appointment Selection Screen



The image shows a mobile application screen for selecting an appointment. At the top, the status bar displays the time 9:41, signal strength, Wi-Fi, and battery icons. Below this is a header bar with a back arrow, the title "Select Appointment", and a menu icon. The main content area is divided into two sections: "Available Dates" and "Available Slots".

Available Dates: This section shows five date buttons for Thursday 20, Friday 21, Saturday 22, Sunday 23, and Monday 24. Friday 21 is the selected date, highlighted in a dark teal color.

Available Slots: This section displays a grid of time slots. The slots are arranged in three rows: the first row has 9:00 AM, 9:30 AM, and 10:00 AM; the second row has 10:30 AM, 11:00 AM, and 2:00 PM; the third row has 2:30 PM and 3:00 PM. The 10:00 AM slot is the selected time, highlighted in a dark teal color. All other slots are in a light gray color.

At the bottom of the screen is a large, dark teal button labeled "Continue".

Available Dates				
Thu 20	Fri 21	Sat 22	Sun 23	Mon 24

Available Slots		
9:00 AM	9:30 AM	10:00 AM
10:30 AM	11:00 AM	2:00 PM
2:30 PM	3:00 PM	

Continue

Figure 7: Booking Confirmation Screen

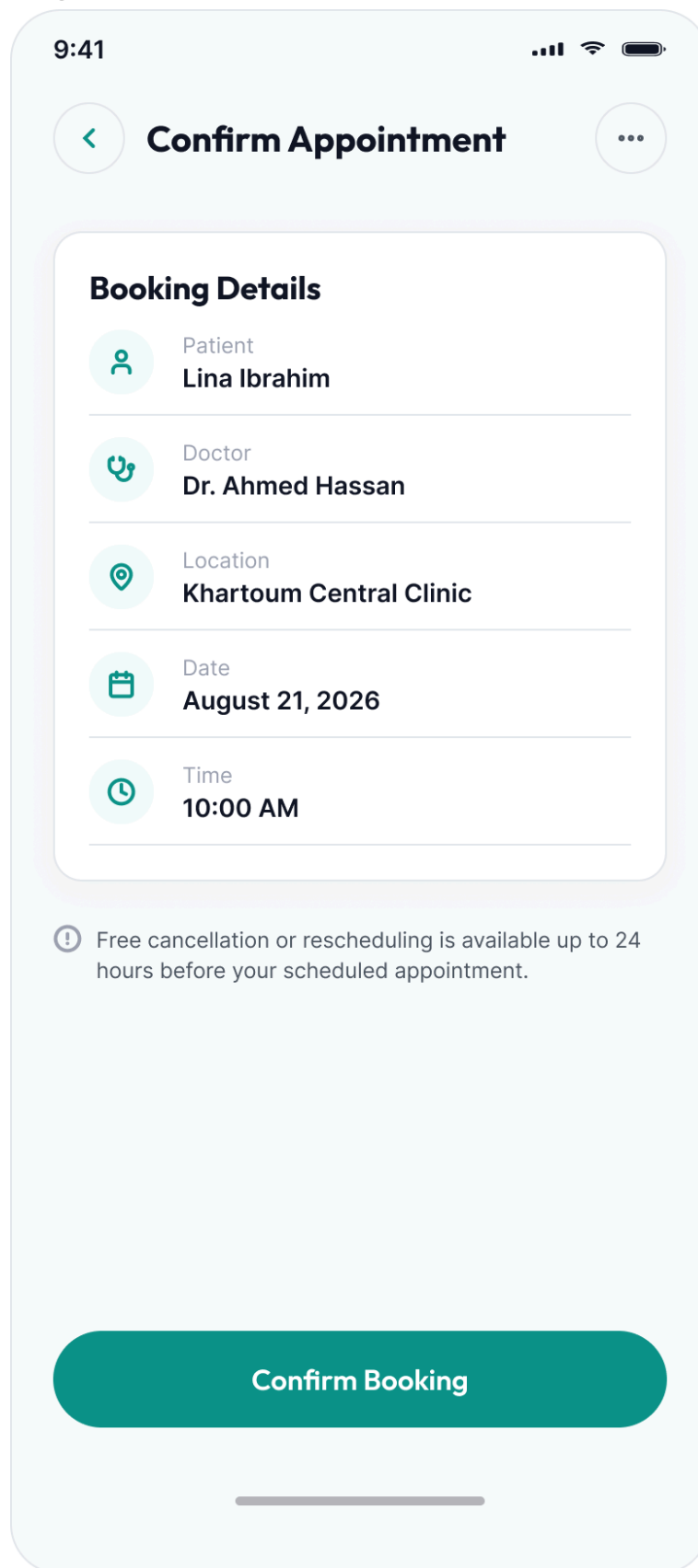


Figure 8: Successful Booking Screen

